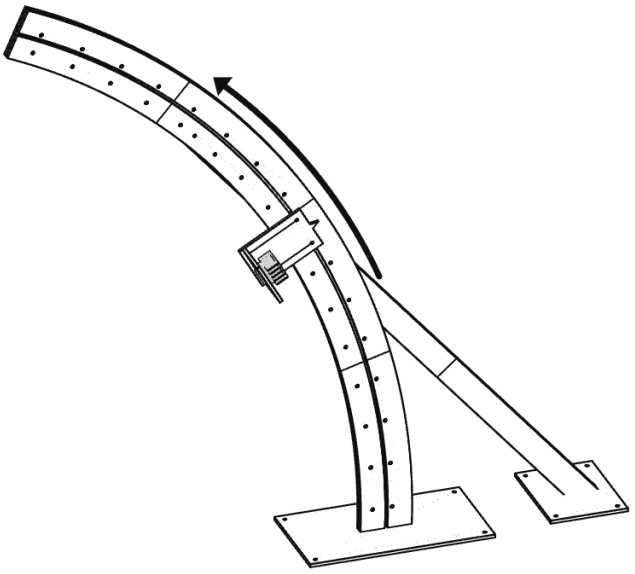
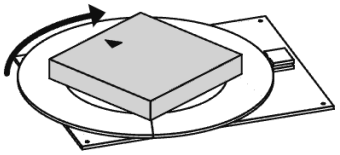
GLENIFF HELIODON

Bench sun simulator · 54.4° N



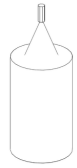


2×

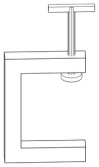
1 night

ATU Sligo · Department of Architecture
ARCT06004 · Architecture Design Studio 1
47 parts · 9 sheets of 6 mm plywood

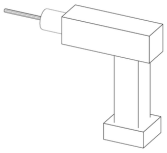
TOOLS



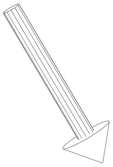
Wood glue



Clamps, 4 or more



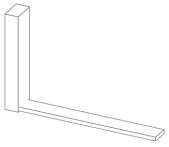
Drill



Countersink bit



Screwdriver

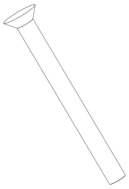


Square and ruler

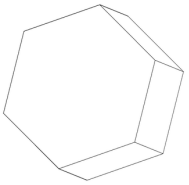
HARDWARE



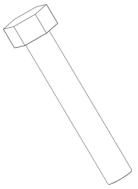
31×
Dowel 6 × 18 mm



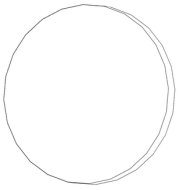
2×
Bolt M5 × 50 countersunk



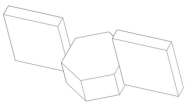
2×
Nut M5



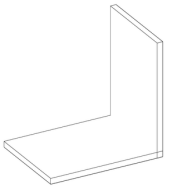
2×
Bolt M5 × 30



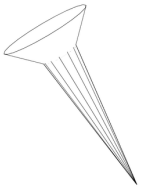
2×
Washer M5



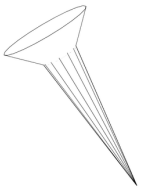
2×
Wing nut M5



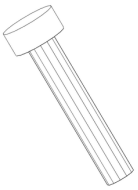
4×
Angle bracket



12×
Wood screw 10 mm



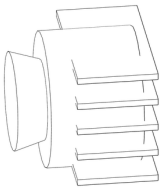
12×
Wood screw 6 mm, for the 6 mm plates



4×
Bolt M3 × 16 + nut



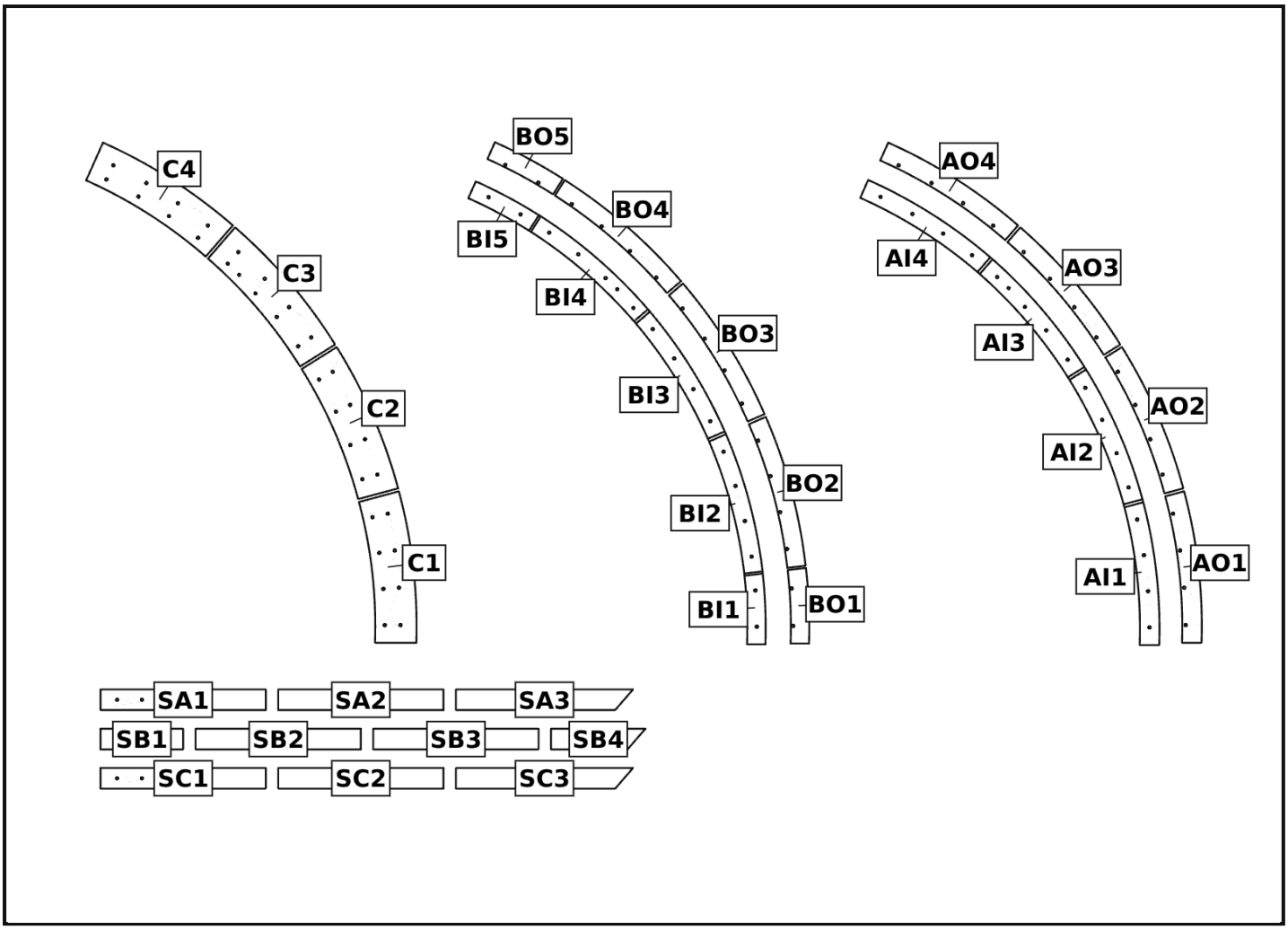
1×
Lazy-susan bearing, 250–300 mm



1×
COB LED 20–30 W, heatsink, lens

Plus: PVA wood glue, 6 mm dowel rod (about 0.6 m, cut into 18 mm lengths), a weight or two for overnight.

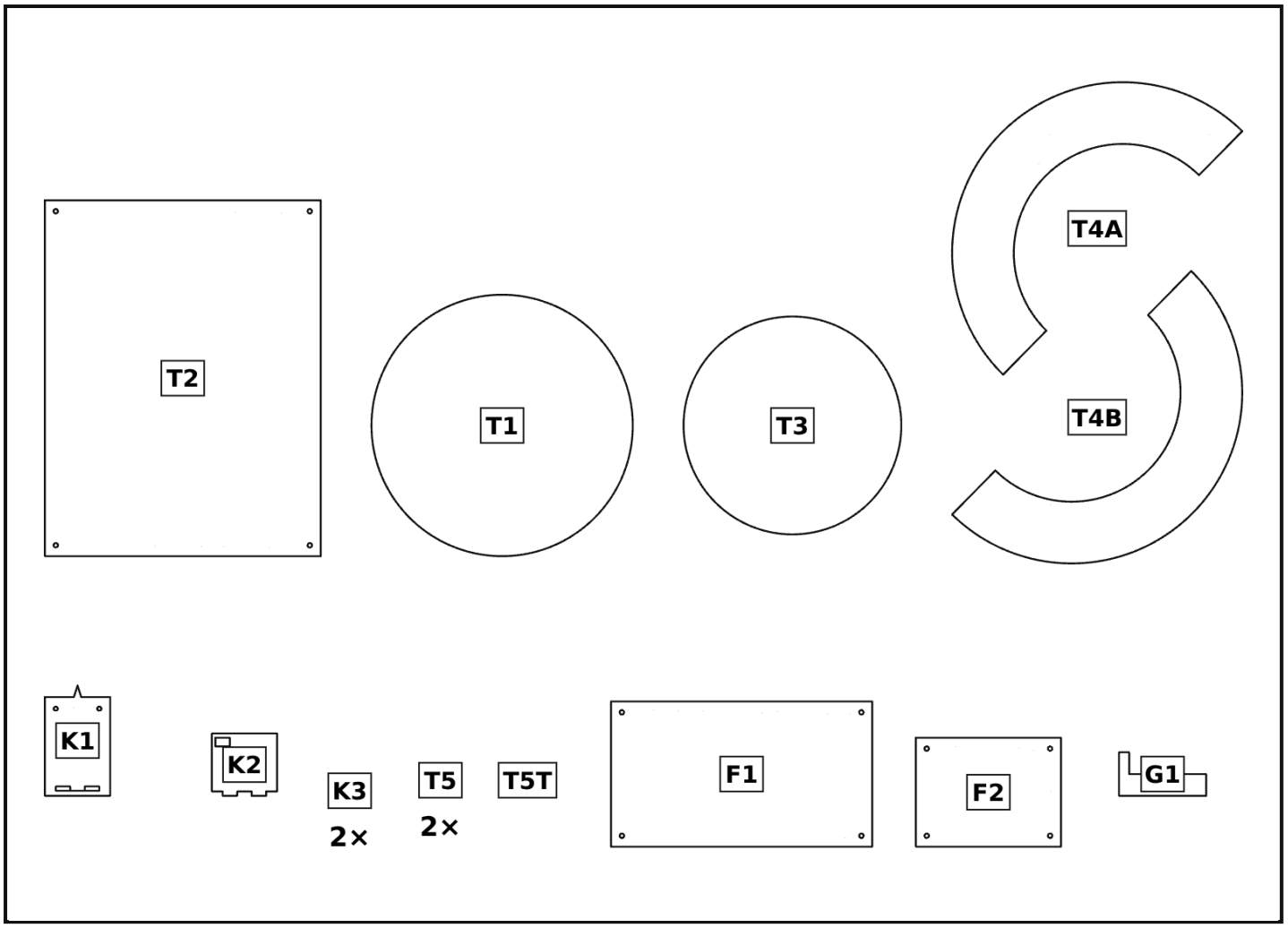
PARTS · ARC AND STRUT



C	**Back layer**, solid. C1–C4. 4 pieces.
B	**Middle layer**. BI inner, BO outer. 10 pieces.
A	**Front layer**. AI inner, AO outer with the scale. 8 pieces.
S	**Strut**, 3 layers. SA, SB, SC. 10 pieces.

Sort the parts into these piles before you start. The codes are engraved on the pieces, except AO1–AO4, which carry the scale numbers instead.

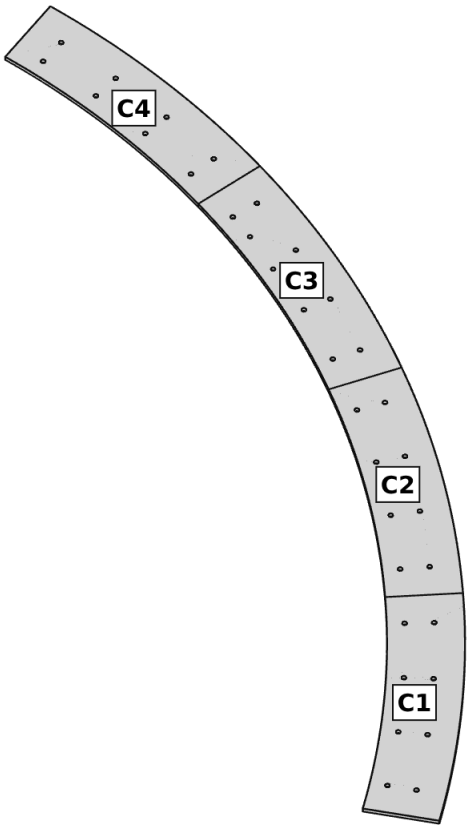
PARTS · TURNTABLE, CARRIAGE, FEET



T	**Turntable**. T1 disc, T2 base, T3 centre, T4A and T4B scale ring, T5 index. 8 pieces.
K	**Carriage**. K1 plate, K2 LED holder, K3 nut plates (optional). 4 pieces.
F	**Feet**. F1 under the arc, F2 under the strut. 2 pieces.
G	**Gauge**. G1 sets the 46 mm gap. Keep it with the rig.

T3, T4A, T4B and the T5 pieces have no code on them: match them by shape.

1

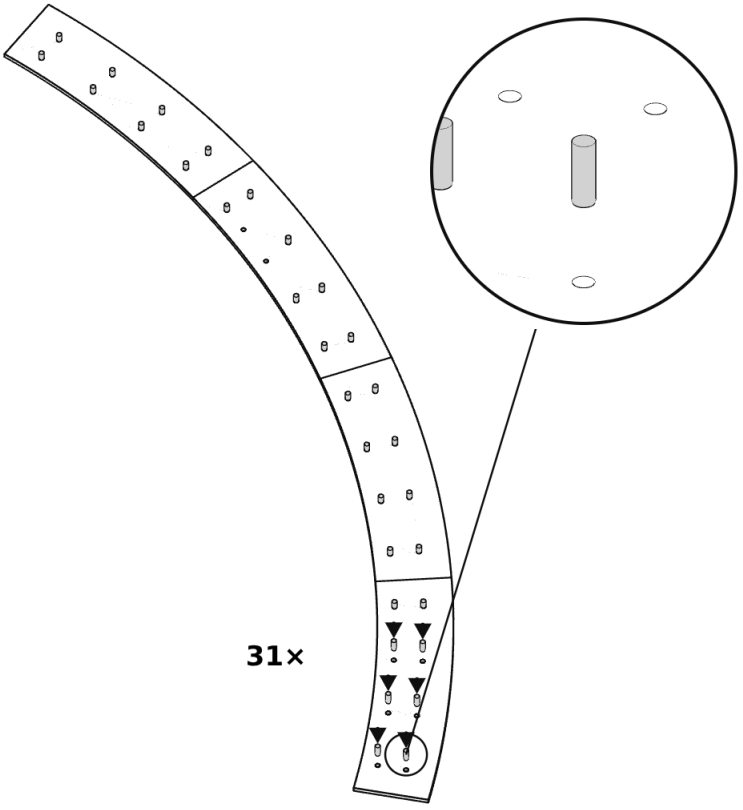


Lay C1–C4 flat on a clean table, engraved side up. Push the joints tight.

2

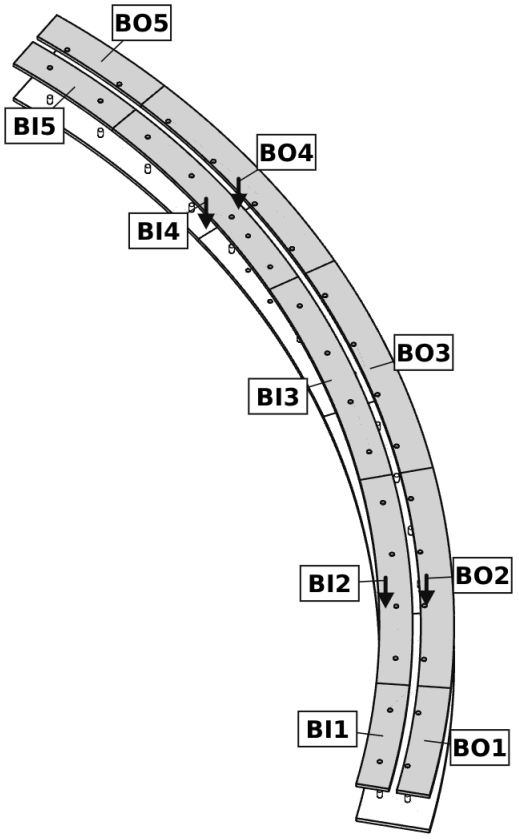


31×



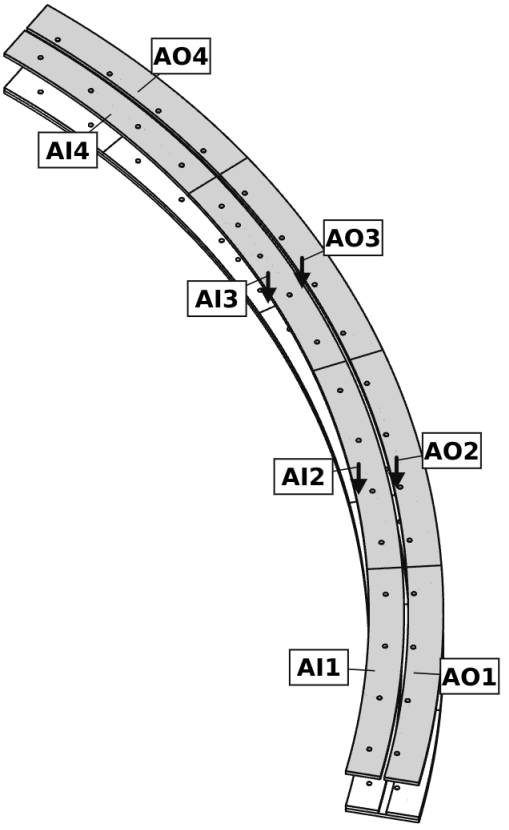
Push a dowel into every hole. Leave the two holes near the 45° mark empty: the strut bolts go there.

3



Glue the middle layer over the dowels: BI inside, BO outside. Leave the 11 mm channel between them open.

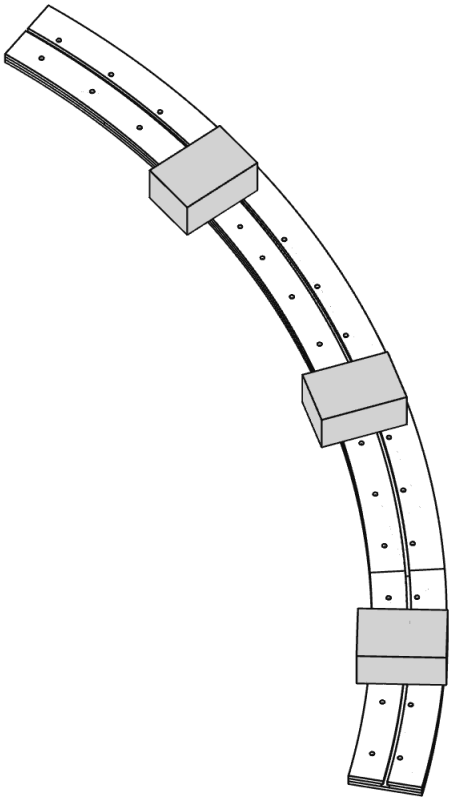
4



Glue the front layer: AI inside, AO outside with the scale numbers up. Leave the 5.5 mm slot open.

5

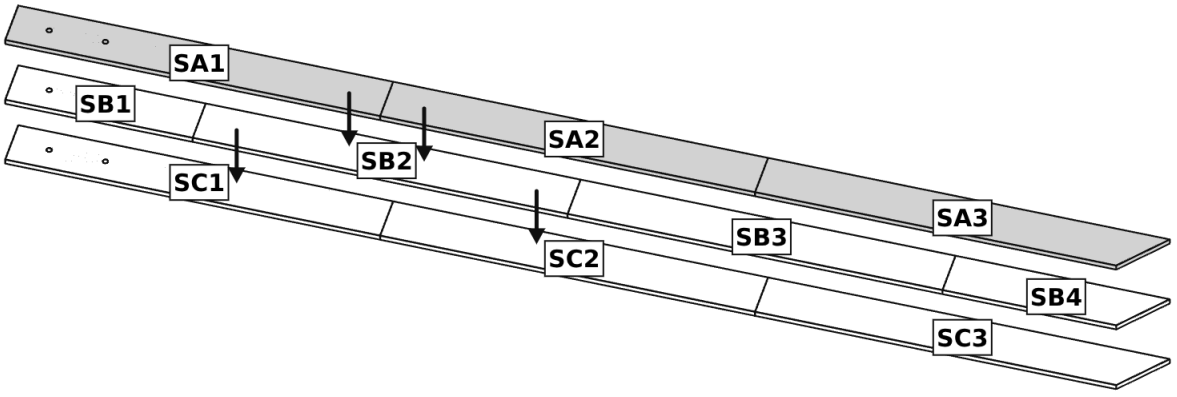
12 h



Weight it flat and leave it overnight.

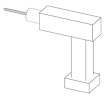
6

4×

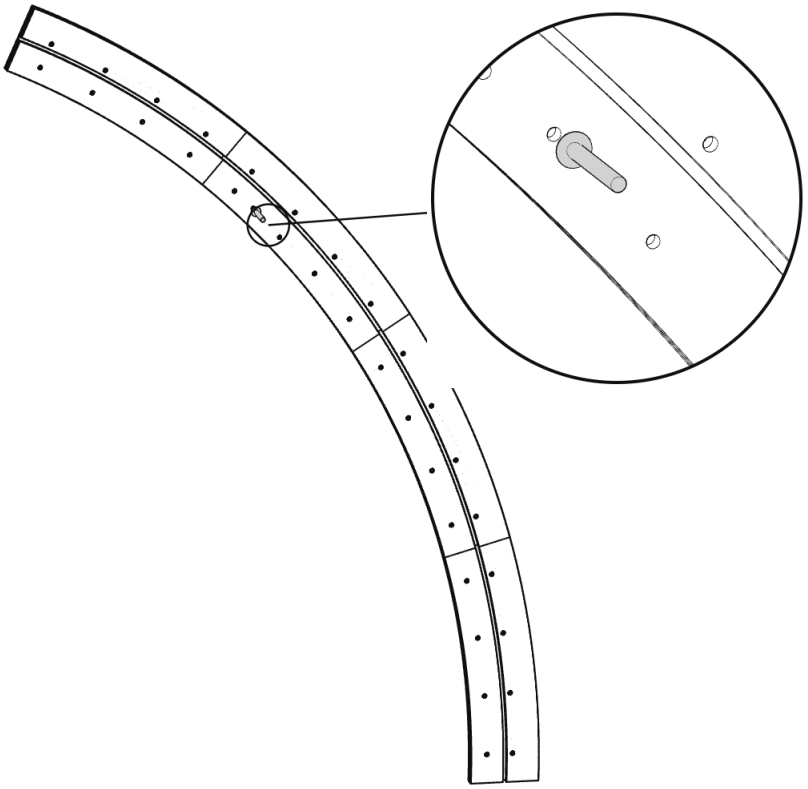


Glue the strut: SC at the bottom, then SB, then SA. Line up the ends; the joints fall in different places. Clamp until dry.

7







Countersink the two strut holes on the front face, so the bolt heads sit flush. The carriage slides over them.

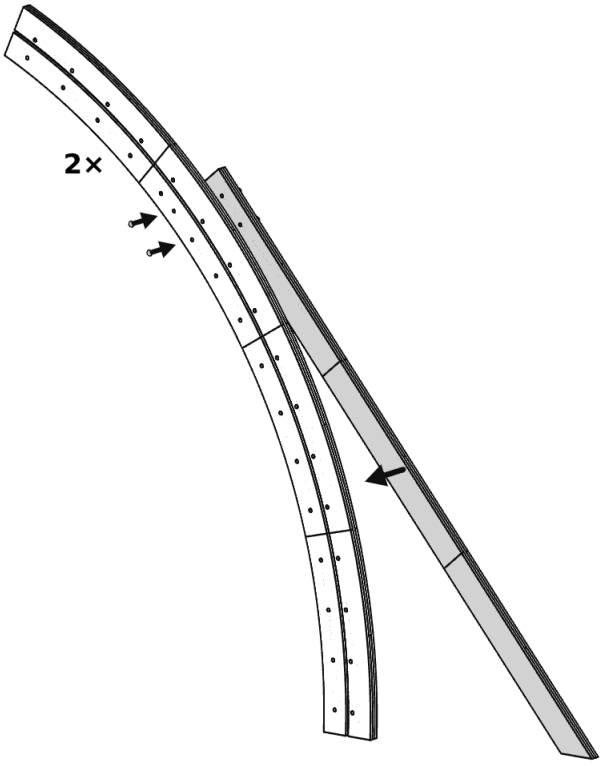
8



2×

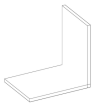


2×



Bolt the strut to the back of the arc, SA side against it. Bolts in from the front, nuts at the back.

9

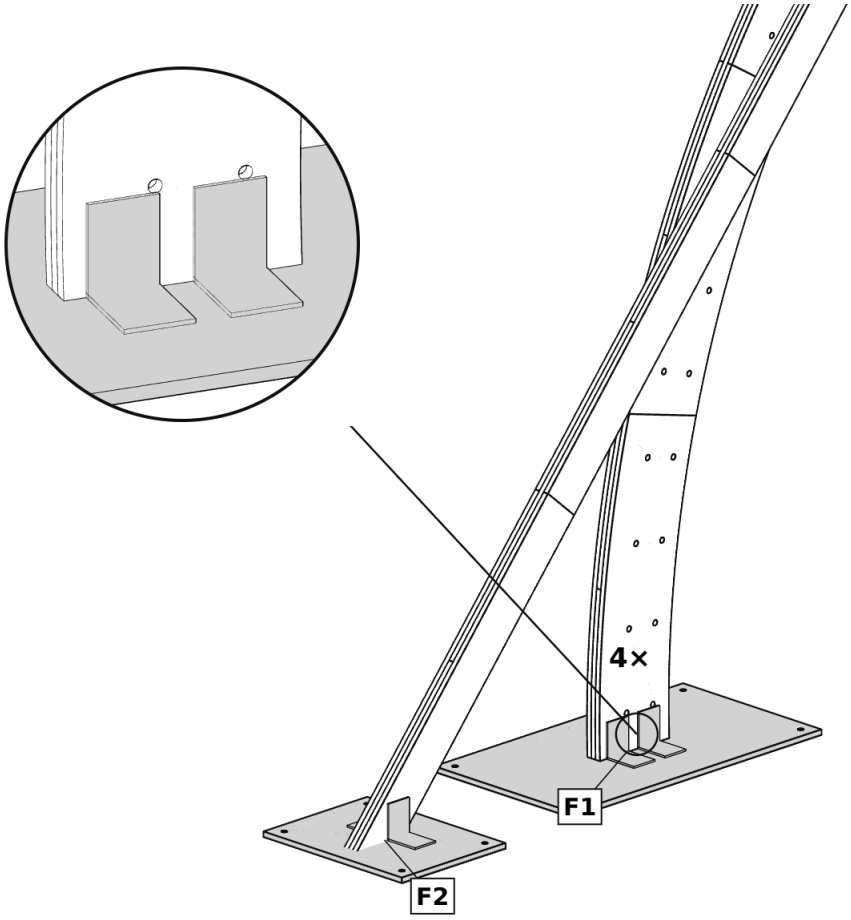


4×



16×



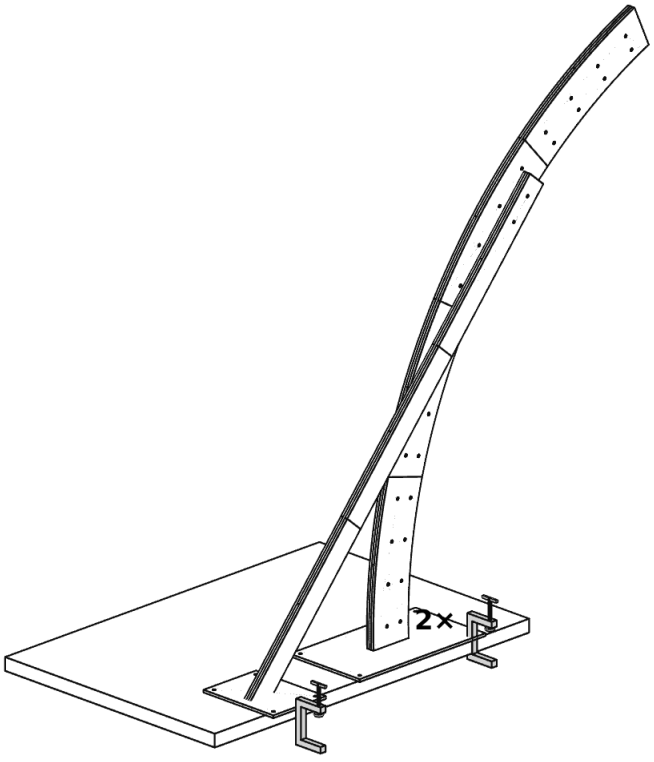


Stand the arc on F1 between the scored lines, and the strut on F2. Fix each with two brackets: 10 mm screws into the arc and strut, 6 mm screws into the feet.

10



2×



Clamp F1 and F2 to the bench so the arc can't move.

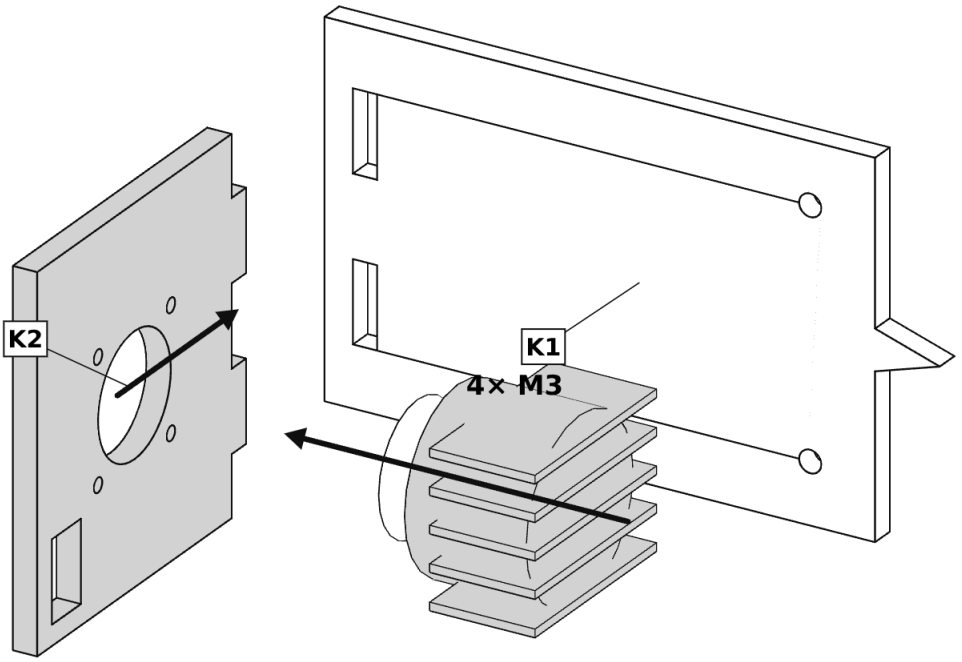
11



4×



1×

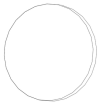


Push K2's tabs into the slots in K1 and glue. Bolt the LED into K2's round hole, lens facing the turntable.

12



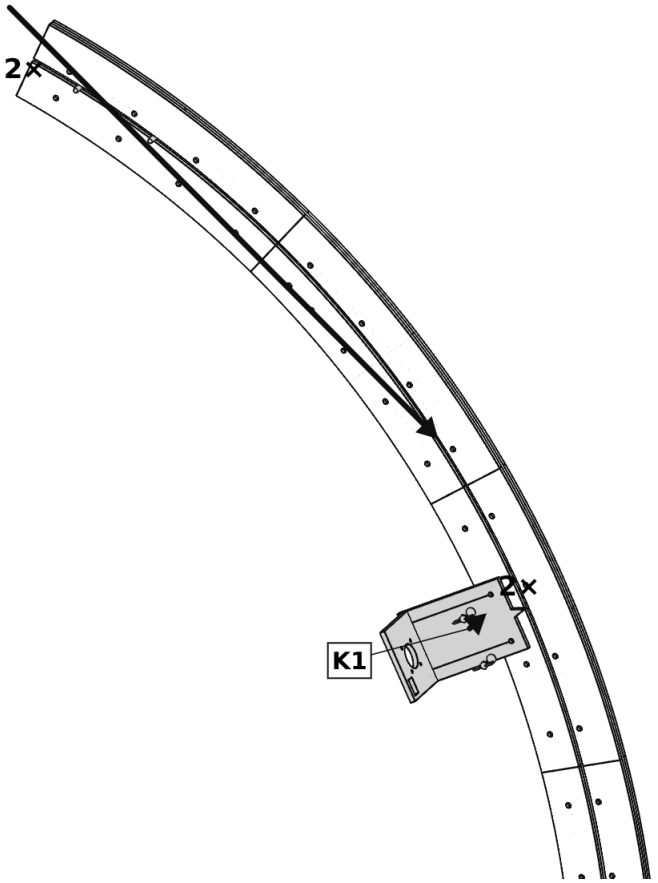
2×



2×

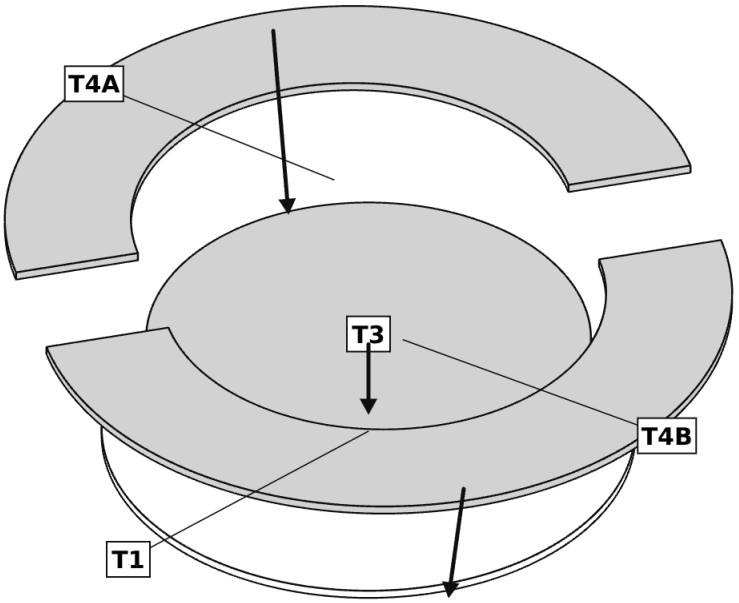


2×



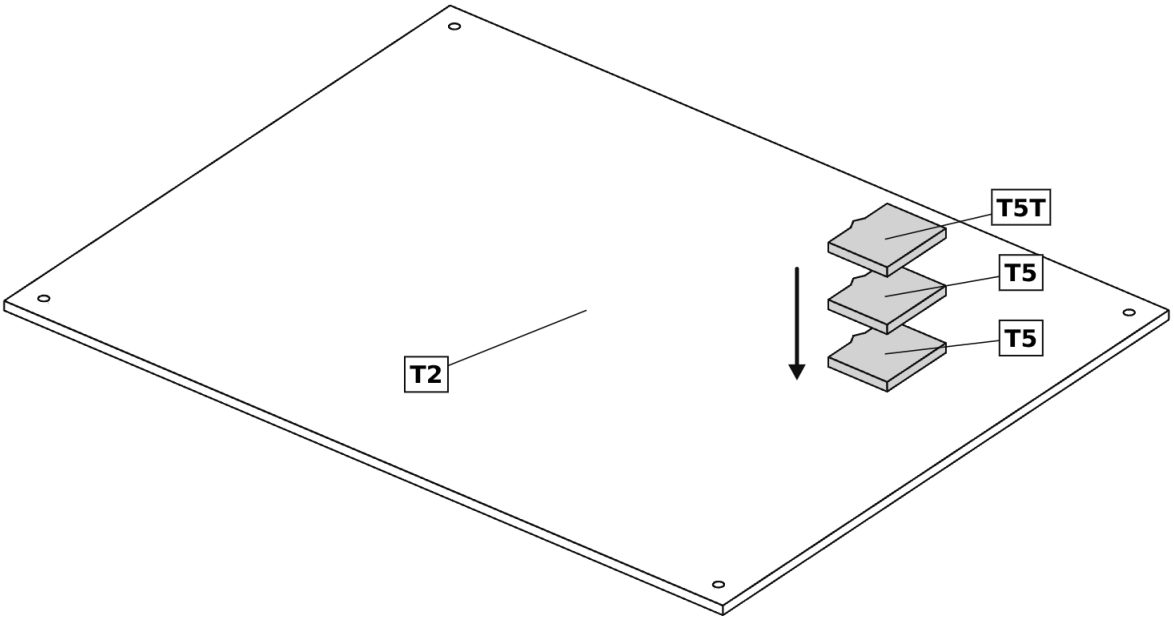
Drop two bolt heads into the channel at the top of the arc and slide them down. Fit K1 over them, then washers and wing nuts.

13



Glue T3 in the middle of T1. Glue T4A and T4B around it, numbers up, with N over T3's MODEL NORTH arrow.

14



Glue T5, T5, then T5T on top, into the scored outline on T2's tab. The tip points to the middle of T2.

15

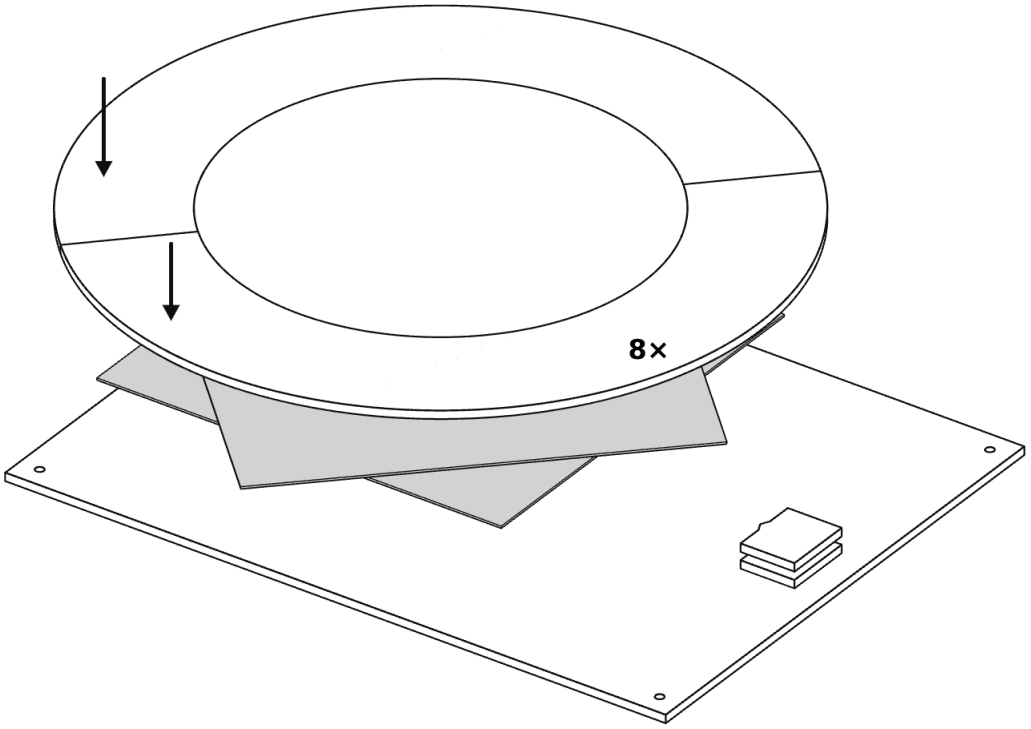


1×



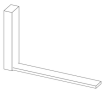
8×

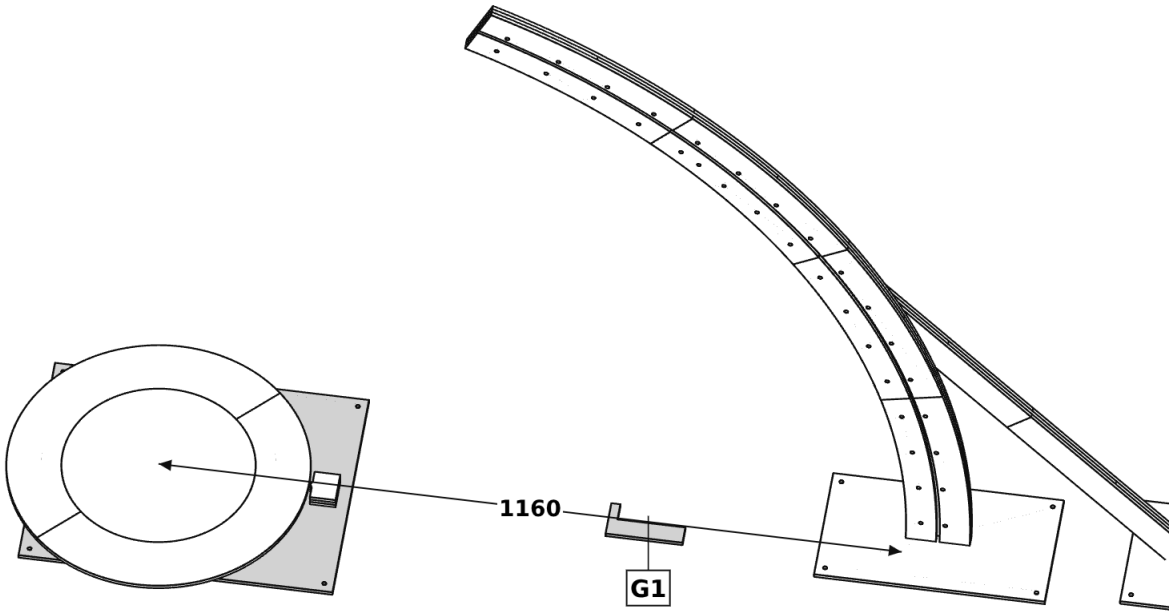




Screw the lazy susan to the middle of T2 with 6 mm screws. Screw the turntable top to it, T1 side down, with 10 mm screws.

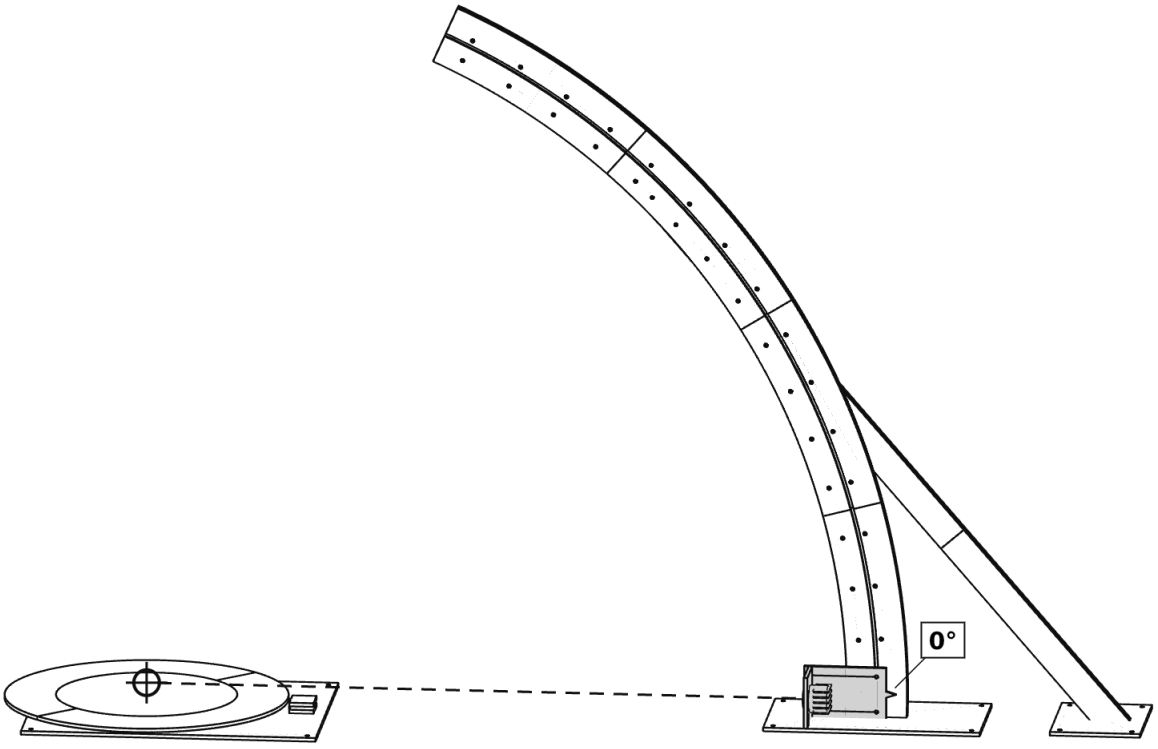
16





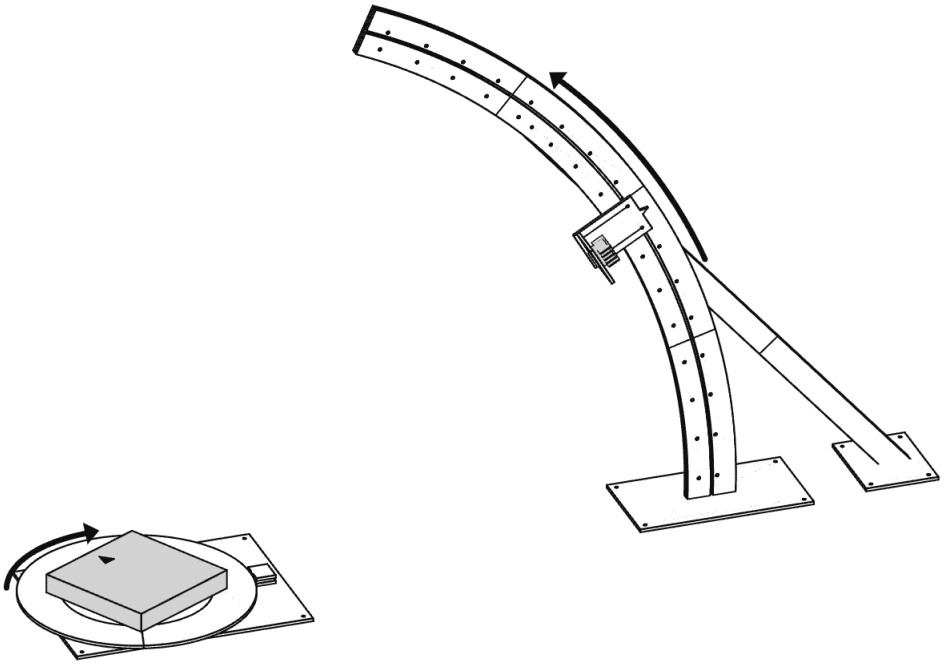
Tab towards the arc. The centre sits 1160 mm from the arc's inner edge, 46 mm in front of its face: set that with G1.

17



Check: slide the LED to 0° and look along it. It should point at the centre of the turntable. Too high or low? Pack under F1 or the turntable.

18



Model north on N. Turn the turntable to the sun's bearing at the index. Slide the LED to the sun's height; tighten the wing nuts.

Sun height and bearing for any date and time: the setting table in the build guide, or the Sun & climate page.